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A study of breast motion using non-linear dynamic FE analysis

Li-Hua Chen^a, Sun-Pui Ng^b, Winnie Yu^{a*}, Jie Zhou^c and K.W. Frances Wan^d

^aCollege of Mechanical Engineering, Beijing University of Technology, Beijing 100124, PR China; ^bInstitute of Textiles and Clothing, The Hong Kong Polytechnic University, Hunghom, Kowloon, Hong Kong; ^cApparel and Art Design College, Xi'an Polytechnic University, Xi'an, China; ^dIndustrial Centre, The Hong Kong Polytechnic University, Hunghom, Kowloon, Hong Kong

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This paper presents a new method to simulate non-linear breast motion by using a three-dimensional (3D) dynamic finite element model (FEM). The model consists of a thorax with two breasts and three skin layers with specific mechanical properties. Using free breast vibration, the viscous damping ratios were ascertained to be 0.215 for an 80B size breast. The shear modulus for the breast was derived as the value that gave the minimum difference between the FEM-predicted results and the experimental data. A hyper-elastic neo-Hookean material model simulated the large deformation of breast tissue. The mode shapes of breast motions at different natural frequencies were established. The highest breast displacement amplitude ratio relative to the thorax was at 4 Hz. The study showed that FEM can predict breast displacement with sufficient accuracy and thereby provide the basis by which bras may be engineered more ergonomically in the future.

Practitioner Summary: To facilitate a theoretical analysis of breast motion to enable the design of more supportive bras, a dynamic FEM based on reliable non-linear properties of breast tissues has been developed. The methods and findings have potential widespread benefit for developing new products to promote women's health and comfort.

Keywords: breast; finite element; dynamic; hyper-elastic behaviour; large deformation

1. Introduction

Excessive breast motion can induce breast pain in women (Page and Steele 1999) and exercise-associated breast pain is positively related to breast displacement (Mason, Page, and Fallon 1999) and fit (White and Scurr 2012; Chen, LaBat, and Bye 2010). Previous analyses of breast displacement were mainly based on empirical studies (White, Scurr, and Smith 2009). Finite element models (FEMs) have been used to predict breast deformations in static states, for applications including breast plastic surgery (Perez del Palomar et al. 2008), clinical biopsy (Azar, Metaxas, and Schnall 2001), modelling of breast compression in X-ray mammography (Ruiter et al. 2002), magnetic resonance mammography and elastography (Washington and Miga 2004). However, FEMs of breast motion and stress distribution in dynamic states are scarce.

Li, Zhang, and Yeung (2003) studied dynamic mechanical bra–breast interactions. A three-dimensional (3D) biomechanical model was built to numerically simulate the deformation and pressure distribution between a bra and two breasts. However, the models were over-simplified by assuming the thorax to be totally rigid with two soft breasts of uniform properties. Yu et al. (2008) developed a FEM of the female foot for high-heeled shoe design to investigate the plantar contact pressure and internal loading responses of the bony and soft tissue structures. However the foot structure is significantly different to breasts.

Biomechanical modelling of breasts using FEMs is still in its infancy (Rajagopal et al. 2008). Three important aspects have yet to be addressed. First, the breast is an inhomogeneous structure that contains different tissue layers. The predominant types of tissues within the breast are fat, glandular tissue and skin (McGhee et al. 2013). They have different mechanical properties that vary according to genetic factors and age (Gefen and Dilmoney 2007). The constitutive material parameters for these tissues are often obtained from *ex vivo* indentation tests (Bakic 2000). Previous biomechanical analyses of static breasts have used a non-linear hyper-elastic neo-Hookean material model for fatty and fibro-glandular tissues and a polynomial model for the skin (Perez del Palomar et al. 2008). However, the effect of the non-linear material properties of the breasts on dynamic breast motion has not been investigated.

Secondly, previous studies mainly used motion experiments to measure breast displacement (Mason et al. 1999; Lorentzen and Lawson 1987) that required a number of female subjects to wear different bras or no bra during physical activities. To recruit a sufficient number of subjects to cover the large variance in breast characteristics is difficult because of concerns and personal embarrassment about performing such activities. If the viscoelastic deformation of various breast

*Corresponding author. Email: winnie.yu.itc@polyu.edu.hk

types can be quantified, then the use of a scientifically validated dynamic FEM to simulate and predict breast motion will obviate the need for the physical participation of women in the experiments, thereby avoiding breast injuries and discomfort.

Thirdly, accurate identification of body parameters for predicting biomechanical responses is very important in FEMs. Tanner, Schnabel, and Hill (2006) found that a model's prediction is highly sensitive to the changes in boundary conditions and material properties. However, the optimum definition of breast boundary and the mechanical properties of breast materials remain unclear.

To address these issues, this work aimed to develop a novel method for building a dynamic FEM to simulate breast displacement during different physical activities, based on individual specific geometric shapes and the non-linear mechanical properties of the breasts. Each breast model, which consisted of a thorax, two breasts and three layers of skin, was characterised by hyper-elastic neo-Hookean material behaviour, to simulate breast displacement and deformation. This study also devised a novel method to measure the damping ratio and predict the shear modulus of individual breast using the motion analysis of breast vibration during free vibration. The results were validated by comparison with experimental motion data.

2. Materials and methods for FEM

2.1 Individual-specific geometric breast model

A healthy Chinese female volunteer with the most prevalent bra cup size 80B was selected to participate in this study. She was 40 years old with tear-shaped asymmetric drooping breasts. She was married and had breast-fed one child 10 years ago. She had a height of 160 cm, weighed 57.8 kg, with a body mass index of 22.6 kg/m^2 . She had not undergone any previous breast surgeries. She gave informed consent to participate in the nude breast scanning and motion experiments. The work was approved by the Human Subjects Ethics Sub-committee of the University.

For 3D breast scanning, commercially available methods are the Voxelan laser scanner (Zheng, Yu, and Fan 2007), TC² white light scanner (Pandarum, Yu, and Hunter 2011) and Intek phase-shifting moiré topography (Lee, Hong, and Kim 2004; Lee and Hong 2007). However, there is no standardised method (Paul and Wischniewski 2012) for digitising human models. In this study, a university-developed 3D photogrammetric body imaging system (Figure 1) was used because high-resolution 3D colour images were essential to form the basis for modelling the breasts to ensure that clear details and landmarks could be identified.

The system comprised an array of digital cameras and flash projectors with their positions fixed to define the scanning space. Twenty Canon PowerShot A650 digital cameras, each with an image resolution of 12 million pixels, were used as the two-dimensional (2D) image capturing devices. Each capturing module comprised two cameras and at least one flash projector. They were concurrently controlled by several computers to achieve the optimum operation speed. This non-invasive optical system acquires high-definition data in terms of geometric and textural information. It generates

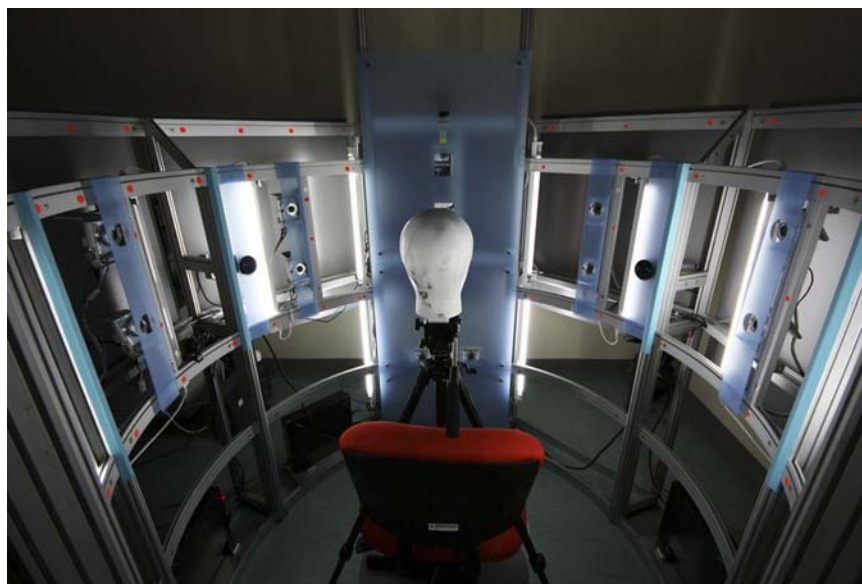


Figure 1. The 3D photogrammetric scanning system.

3D high-resolution point clouds with an average point-to-point distance of less than 1 mm. The level of details to be captured can be adjusted through various zoom-lens settings of the cameras. Unlike other commercial body scanners, this system allows very fine surface features to be captured. The 3D surface of the breast object was built by registering the object features in the 2D images. Both the surface data and the corresponding colour textural information of the object were obtained.

Before capturing the breast images, six black spots were marked on the skin surface of the subject's breast (Figure 2a). The locations of these spots were the same as the marker points used in the breast motion experiments (Section 3), so that the model-simulated results could be compared with the corresponding points in the experimental results.

A geometric model of the breast surface was constructed and is shown in Figure 2b. Identification of the breast boundary was based on the surface measurements from the nipple to body landmarks on the breast boundary. Female breasts cover the chest area between the midaxillary line and the edge of the sternum, overlaying the anterior serratus and pectoralis major muscles from the second rib to the sixth or seventh intercostal cartilage. The upper and lateral portions of the breast, known as the tail of Spence, extend into the axillary region and pass through an opening in the pectoral fascia, the foramen of Langer, into the axilla (Walker, Hall, and Hurst 1990). Based on this definition of the breast, the landmarks on the breast boundary were identified by the same researcher through direct observations.

2.2 Mechanical properties used in the breast model

The breast model consisted of a thorax, two breasts and three-layers of skin with different mechanical properties. In humans, the bony framework of the thorax consists of the 12 thoracic vertebrae, 12 pairs of ribs and the sternum: it contains the lungs, air passages, heart and vessels bounded by the diaphragm. The bony framework is encased with muscles, fat and subcutaneous tissues. In this study, the thorax was assumed to be uniform in density and modulus. This was because increasing the number of degrees of freedom in the thorax would unnecessarily complicate the FEM and increase the computational time. The findings from this study would also enable the validity of these assumptions to be evaluated. According to Zheng (2007), the thorax may simply be considered as a linearly elastic material but with a very high value of Young's modulus (489 kPa) relative to the values for skin (0.17–18.8 kPa) and breasts (0.34 kPa) as shown in Table 1.

The skin was treated as a three-layer shell element that contained the epidermis, dermis and hypodermis (Gefen and Dilmoney 2007). The thickness of the epidermis ranges from 50 to 100 μm . The dermis is mainly composed of collagen and elastic fibres embedded in a viscous medium made of water and glycoproteins. The total thickness of the dermis varies from 1 to 3 mm. The hypodermis primarily consists of loose connective tissues and lobules of fat. Therefore in this study, the thickness was set to 0.1 mm for the epidermis, 1 mm for the dermis and 1 mm for the hypodermis. Human skin mechanical properties vary substantially with age, hydration, obesity, disease and anatomical sites. The neo-Hookean model considers the skin as hyper-elastic and its material parameters would greatly influence breast deformation and motion. Based on Hendriks et al. (2003), the assumed density and Poisson's ratio of skin are specified in Table 1. The reason for using three layers of skin was to simulate the dynamic behaviour of human skin more realistically. The three layer model is an improvement over a single layer model because it avoids the potential errors from the homogenising programme that combines the three skin layers into a homogenous material parameter.

Breast tissues are inherently viscoelastic in nature as they exhibit both viscous and elastic responses (Sinkus et al. 2005) to dynamic loading. These responses depend on the strain rate or history of strain. In the case of static and quasi-static problems, the forces in the breast are dominated by its elastic response, and the breast tissues can be primarily modelled as

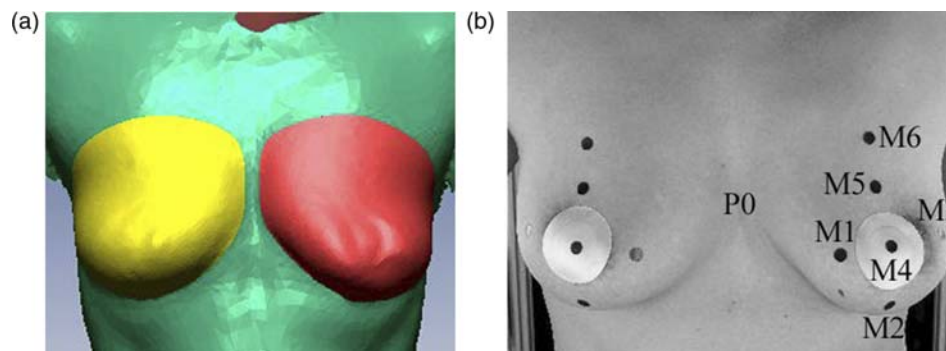


Figure 2. Development of a geometric breast model in standing position (a) reference markers and (b) the breast model.

Table 1. Material parameters input for the FE breast model.

Components		Parameter	Value
Thorax		Density (kg/m ³)	1579
		Young's modulus (kPa)	489
		Poisson's ratio	0.3
Breast		Density (kg/m ³)	1000
		Shear modulus (kPa)	0.34
		Poisson's ratio	0.495
Skin	Hypodermis	Density (kg/m ³)	970
		Shear modulus (kPa)	0.17
		Poisson's ratio	0.495
	Dermis	Density (kg/m ³)	1100
		Shear modulus (kPa)	18.8
		Poisson's ratio	0.495
	Epidermis	Density (kg/m ³)	1000
		Shear modulus (kPa)	0.35
		Poisson's ratio	0.495

elastic. However, the dynamic behaviour of the breast during physical activities can also be affected by its inherent viscous property. Therefore, it was necessary to devise a method to measure the viscous damping ratio and predict the shear modulus of an individual breast.

In this work, FEM software ANSYS 12.0 (ANSYS Inc., PA, USA) was used to simulate breast motion and a classical standard Voigt model was used to characterise the viscoelastic deformation of the breast tissues. The overall stress tensor for a viscoelastic material is divided into elastic and viscous components. The elastic component, expressed in terms of shear modulus G , was assumed to be isotropically hyper-elastic and quasi-incompressible with a Poisson's ratio of 0.5. The viscous component is normally expressed in terms of viscosity constant η . However, instead of η , the Voigt model needs the input of a viscous constant damping ratio, ζ , which is the ratio of the viscous parameter to the critical damping parameter. The damping ratio of a vibration system can be measured based on the characterisation of damping, such as half-power bandwidth, quality factor, loss factor and logarithmic decrement. The logarithmic decrement is measured in the time domain (Bishop and Johnson 1979), not in the frequency domain, so this approach was applied in this study. The free-vibration displacement amplitude history of a system to an impulse excitation is measured and recorded. The logarithmic decrement refers to the natural logarithmic value of the ratio of two adjacent peak values of vibration displacement in free decay.

A free-vibration test of the volunteer's breast was performed with the subject in a standing position. A small spherical retro-reflective marker was placed on the nipple of the left breast. When the participant raised her breast vertically by hand and then quickly released it, the breast began to decrementally freely vibrate. The time-displacement curve is shown in Figure 3, which exhibits the typical feature of amplitude attenuation due to viscous damping.

As a deformable continuum, the breast has an infinite number of natural frequencies and mode shapes. The logarithmic decrement method was used to obtain the value of the viscous damping ratio in Figure 3. The logarithmic decrement δ is

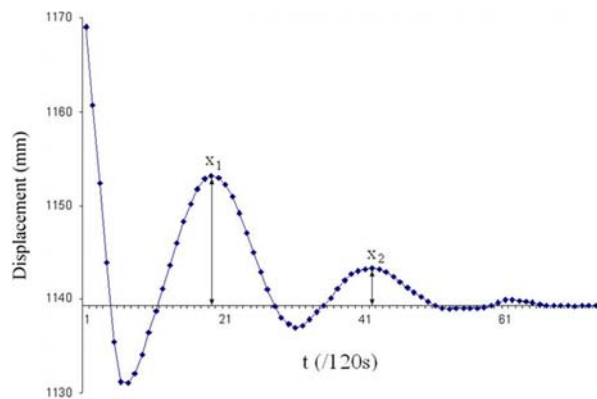


Figure 3. Displacement-time curve of breast indicating the motion decay for viscous material.

expressed by Equation (1).

$$\delta = \ln \left(\frac{x_n}{x_{n+1}} \right), \quad (1)$$

where x_n is defined in the displacement–time curve of motion decay of the breast (Figure 3). Normally $n = 1$ or 2 .

The damping ratio ζ is expressed in terms of the logarithmic decrement δ .

$$\zeta = \frac{\delta}{\sqrt{(2\pi)^2 + \delta^2}}. \quad (2)$$

By lifting the breast to different initial heights, 10 experiments of free vibrations were conducted. From this, the mean value of viscous damping ratio ζ of the subject was finally ascertained to be 0.215 ± 0.013 .

For ease of modelling and computation, the various tissues contained in the breast were not considered and an average value of the shear modulus G was initially assigned to all of the breast tissue. The breast was assumed to be incompressible, homogeneous and isotropic, so its elasticity could be characterised by neo-Hookean material behaviour (Perez del Palomar et al. 2008). The strain energy density W , as a function of quasi-incompressible hyper-elastic materials in terms of the strain invariants, can be written as:

$$W = \frac{G}{2} \cdot (\mathbf{I}_1 - 3), \quad (3)$$

where G and $\mathbf{I}_1$ are the shear modulus and strain, respectively.

$$\mathbf{I}_1 = \text{trace}(\mathbf{F}^T \mathbf{F}), \quad (4)$$

where $\mathbf{F}$ is the deformation gradient.

According to Katch et al. (1980), breast tissue density is assumed to be close to that of water. Breast tissue is heterogeneous and contains many different kinds of tissue such as fat and glandular tissue. However, there is limited information to identify the mechanical properties of the breast. For simplicity, the breast density was considered to be uniform and similar to water with an assumed variation within 5% of the bulk tissue density (1000 kg/m^3), and the shear modulus of the individual breast was derived through iterative computation and incrementally refining the initially assigned value until the minimum difference between the FE predicted data and the experimental motion data was obtained.

Healthy breast tissues appear to be viscoelastic but vary among individuals due to age, parenchyma and menstrual days. The *ex vivo* experimental parameters of breast tissue material given in the literature are unreliable. The stress–strain laws proposed for fatty and fibro-glandular tissues vary hugely. Even measurements of the shear modulus of breast tissues could vary (Gefen and Dilmoney 2007). Therefore, the value of the shear modulus is better estimated using FEM in addition to *in vivo* experimental data. First, the initial value of shear modulus G was assigned to be 0.1 kPa and then increased gradually. The quality of fit between the predicted and experimental data of breast displacement was evaluated in terms of the root mean square error (RMS) averaged across all the sample data points and marker points in the x , y and z directions for three activities *viz.* walking, running and stepping performed by each participant. Equation (5) defines the RMS,

$$\text{RMS} = \frac{100}{N_t \cdot N_d} \sqrt{\sum_{i=1}^{N_d} \sum_{j=1}^{N_t} \left(\frac{D_{ij}^E - D_{ij}^F}{D_{ij}^E} \right)^2}, \quad (5)$$

where N_t is the number of sample data points, N_d is the number of markers, D^E and D^F are the displacements obtained from the experiments and FEM, respectively. When the function satisfies the criterion $\text{RMS} < 1\%$, the corresponding constant values of the shear modulus of the breast and skin were noted. These are shown in Table 1.

2.3 Mesh creation

A geometric breast model was generated for the subject using the FEM software ANSYS 12.0 (ANSYS Inc., PA, USA). A 3D 10-node tetrahedral solid element which has quadratic displacement behaviour and is well suited for the modelling of irregular meshes was used to model the breast and thorax. The skin was defined as a surface coating based on the solid volume of the breasts and thorax. This coating was meshed by using three-layer shell elements, which represented the

epidermis, dermis and hypodermis, each with different material parameters and thicknesses. These nodes which formed a 2D mesh of skin were the same nodes that corresponded to the solid element at the borders of the breast and thorax model. The average total element number of the body was 28,972 (9457 solid elements for the thorax, 16,006 solid elements for the breast and 3509 shell elements for skin). The average of the total number of nodes was 35,032.

2.4 Loading and boundary conditions

The finite element (FE) breast model was constructed for the participant in a standing configuration. The model was considered as a reference for the static equilibrium position of the breast, to which no external force was applied. The initial breast deformation due to gravity was eliminated (Thomson and Dahleh 1998), but the inertia force from the breast mass was calculated in the dynamic analysis of loading and displacement. For the analysis of breast motion, the standing configuration in an unloaded state provided a better approximation than a prone configuration. It is acknowledged that this did not represent the true stress-free unloaded state because gravity loading exists.

As breast tissues are attached to the chest muscles on the ribcage, displacement boundary conditions were applied at the posterior surface of the breast. The thorax was assumed to be almost rigid (linearly elastic but with a high Young's modulus compared to the skin and breast tissues) in the biomechanical model. The boundary displacements were regarded as the displacement of the thorax.

3. Experimental

The FE model was used to simulate the 3D dynamic breast displacement during three types of activities. In order to evaluate its feasibility and accuracy, breast displacement was simulated by each respective FE model and then compared with the data obtained from the motion experiments. The breast displacements relative to the ground and the thorax were both calculated.

The FE model was evaluated by experiments that were carried out in the conditioned Human Locomotion Laboratory at the University. The bare-breasted subject was instructed to perform three types of activities: first, walking on a treadmill; second, slowly running on a treadmill and finally, stepping up and down on a 24-cm high platform at a predetermined speed. The 3D body kinematics was recorded using a six-camera Vicon motion capturing system (Pulnix 6701AN, Oxford Metrics, UK). The dynamic 3D coordinates of specific breast markers were recorded at a 120 Hz sampling frequency. Both static and dynamic calibration procedures of the motion capturing system were carried out before the experiments, with residual of less than 1 mm.

Small spherical retro-reflective markers (9.5 mm in diameter and 1.81 g in mass) were placed at seven different points on the breast skin surface of the subject, as shown in Figure 2a. The thoracic reference marker (P_0) at the front processus xiphoideus was used as the base point for calculating the breast displacement relative to the thorax. It is acknowledged that the skin movement during postural change may potentially cause errors, but these were likely to be very small. The markers M1, M2, M3 and M5 are 4 cm apart from the nipple (M4) in the horizontal and vertical planes. M6 is 4 cm apart from M5. Displacement of these six breast markers was obtained relative to the ground.

The displacement of P_0 was regarded as the displacement of the thorax, which was the displacement boundary condition of the dynamic FEM. The FEM-predicted displacements of these six marker points were then compared with the corresponding markers in the experimental results.

4. Results and discussion

4.1 Model validation

As the mean difference in breast displacement within all the markers only ranged from 0.8 to 2.5 mm, only M2 (4 cm below the nipple) has been presented as a representative example. Figure 4 shows the comparison between the predicted results and experimental data for the displacement of the breast and thorax during (a) walking, (b) running and (c) stepping for the subject in the x , y and z directions in the global coordinate system. It shows that the simulated results in the y and z directions matched well with the experimental results. The simulated displacement in the z direction compared favourably with the actual displacement in terms of frequency and amplitude. Table 2 shows that the model was able to predict the breast displacement in which large deformation and non-linear material parameters were involved with good accuracy.

Relative to the ground, the actual thorax displacements and the breast displacement were almost identical, because their relative displacement was too small to be visible in the global coordinate system.

The mean values of RMS averaged across all markers are shown in Table 2. The RMS error was very small except for running in the x direction. This was probably due to the assumptions of an almost inflexible thorax and simplified boundary conditions in the model. It is known that the arms swing left and right during exercise, which results in shoulder rotation, so

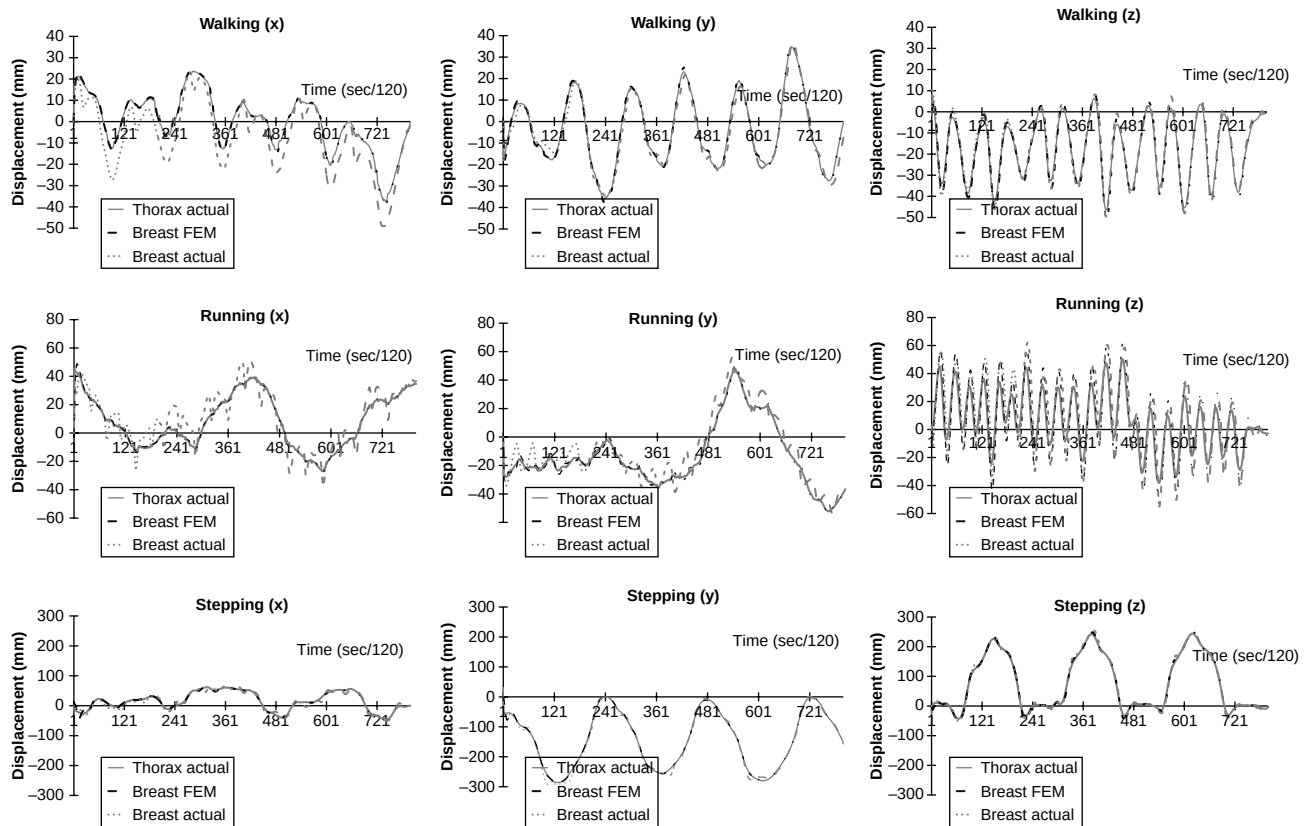


Figure 4. Predicted and actual displacements of the breast and the thorax during (a) walking, (b) running and (c) stepping in x , y and z directions.

Table 2. RMS error percentage of the predicted breast displacement in the x (medial–lateral), y (anterior–posterior) and z (posterior–inferior) directions.

RMS%	x	y	z
Walking	0.27	0.09	0.18
Running	1.59	0.65	0.65
Stepping	0.2	0.05	0.09

the thorax may be more flexible than assumed. There is relatively less movement in shoulder rotation than vertical body movements, so this discrepancy would not affect the overall validity of the FE model. The results indicate that the models may need to be refined to allow for excessive arm swinging movements by some subjects whilst performing some activities. In the more dynamic cases, the lower portion of the breasts motion is restricted due to contact with the chest wall. Without imposing such a self contact constraint within the model, it may be that the breast motion in this region is exaggerated in the model.

4.2 Modal analysis

To interpret the breast motion data, it was necessary to study the characteristics of the breast vibrations. People usually walk or run with a step frequency that ranges from 1.5 to 4.5 Hz. Therefore, the first four natural frequencies and mode shapes were of primary interest. Based on the results of the free-vibration tests, the first 10 natural frequencies of the breast estimated by the FE model with and without viscous damping are shown in Table 3.

For free vibrations, only two mediolateral modes (at 4.87 Hz and 4.91 Hz) and two vertical modes (at 4.99 Hz and 5.02 Hz) were found within a range of 0–5 Hz. The vibration mode shapes of the first mediolateral and vertical modes of the breast are shown in Figure 5.

Table 3. The first 10 natural frequencies of subject A's breast determined with and without considering damping effect.

	With viscous damping	No viscous damping
1	4.87	5.01
2	4.91	5.08
3	4.99	5.13
4	5.02	5.19
5	6.63	7.03
6	6.77	7.20
7	6.91	7.36
8	7.10	7.61
9	7.41	7.99
10	7.69	8.37

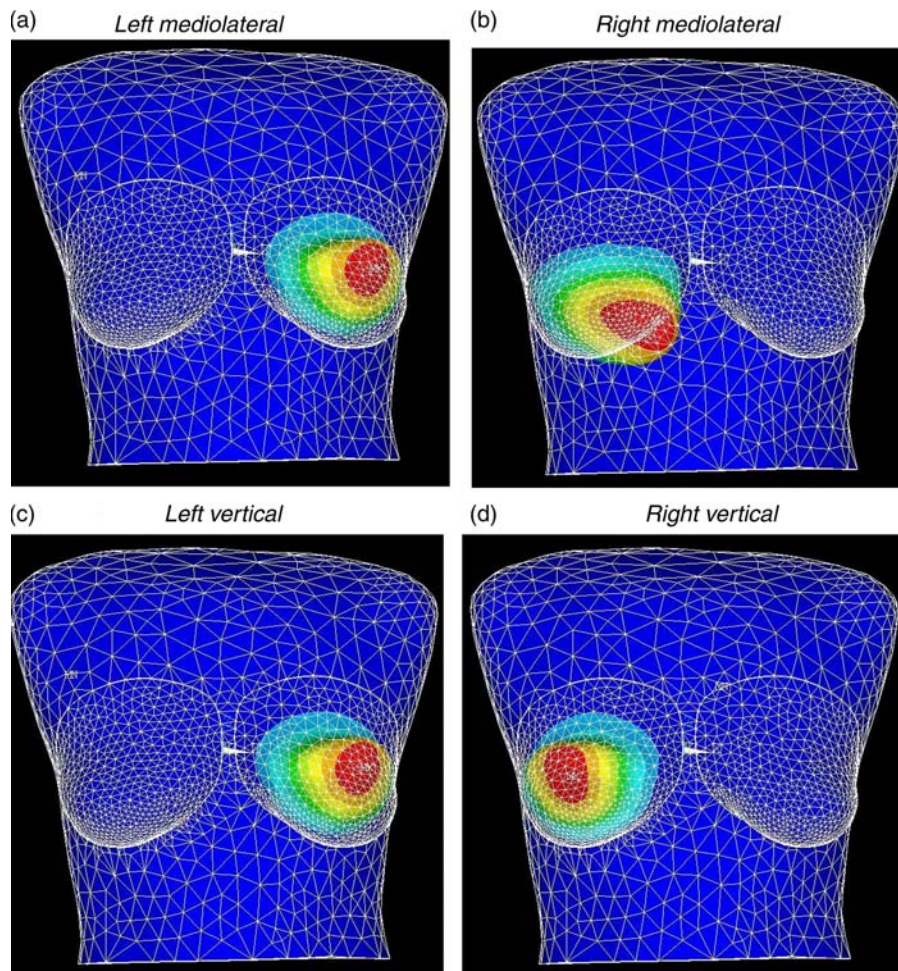


Figure 5. The mode shapes of the breasts: (a) left mediolateral, (b) right mediolateral, (c) left vertical and (d) right vertical. (x: medial–lateral, y: anterior–posterior and z: superior–inferior).

The size and shape of female breasts vary. Differences in breast size are attributed to variations in the amount of adipose tissue. The asymmetric nature of the breasts may explain a difference in the natural frequency between the left and right breasts, whereby the larger breast usually has a lower natural frequency than the smaller breast.

The feasibility and accuracy of the FE results on the vibration characteristics were tested with breast motion experiments at various walking, running and stepping speeds (1.5 ~ 4.0 Hz with an increasing interval of 1/6 Hz). Figure 6

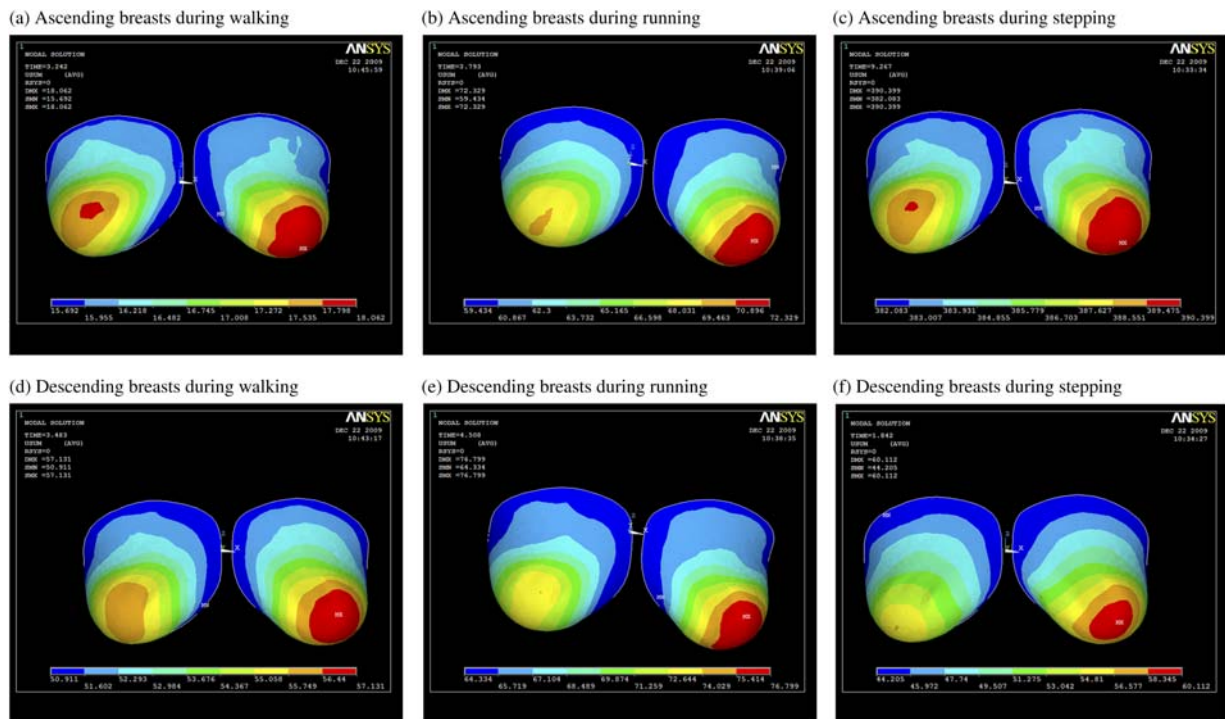


Figure 6. Dynamic deformation simulated by FEM of the ascending breasts during (a) walking, (b) running and (c) stepping; and descending breasts during (d) walking, (e) running and (f) stepping.

shows the dynamic deformation of the ascending and descending breasts simulated by the new FE breast skin model. The predicted outcomes gave confidence in the validity of the values used. Further modelling and analysis would be needed to evaluate whether treating the skin as three separate layers rather than a homogeneous body is necessary or justified. Sensitivity analysis can be performed to determine how sensitive the predictions were to the changes in the values of the material parameters. However, the newly developed FE skin model showed a promising approach for biomechanical modelling. Future researchers can directly apply this new model to more complicated breast modelling with new considerations of the fat and fibroglandular tissue.

Based on the experimental data, an amplitude–frequency characteristic curve was derived and is shown in Figure 7, where the step amplitude ratio is the amplitude of the breast motion at M2 divided by the amplitude of the thorax vibration

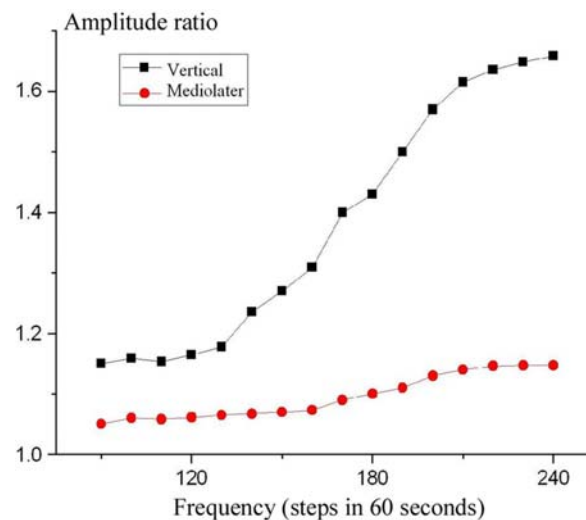


Figure 7. A plot of amplitude ratio of breast against the step frequency.

at P_0 . It shows that the highest amplitude ratio in the vertical direction occurred at a step frequency of 4 Hz, which was close to the first FEM-predicted natural vertical frequency of the breast (Table 3).

Unlike previous models in the literature, the model used realistic viscous non-linear materials to simulate human breasts. The dynamic modal analysis was validated against experimental data and was shown to be valid under the assumption that the thorax had some flexibility but with relatively little shoulder rotation. If a woman runs with excessive shoulder rotation, the breast motion in the x direction cannot be ignored. For further studies, the effects of shoulder rotation on breast motion should be considered. It is acknowledged that the breast material is simplified in this study. For medical applications, MRI scan images of individual breast could be used to build a body geometrical and mechanical model based on the distribution of fat and fibroglandular tissue within the breast. However, this would be far too expensive and inconvenient for general applications. In future studies, more human subjects will be needed for enhancing and further evaluating the accuracy of the FE model. However, the specific density of an individual breast can also be calculated by the weighted percentage of fat and fibroglandular tissue within each breast, which may be predictable by age (Lee et al. 1997).

The model has afforded an improved understanding of breast motion and provides the belief that it can offer the basis by which breast displacement may be predicted during physical activity. Further work is necessary to solve the most difficult problem which is the contact mechanics between elastic bra materials and viscoelastic breast tissues to simulate the dynamic displacement arising from different bra fittings for different breast groups. However, on this evidence, FEM can be a valuable tool for future research to solve problems such as multi-axial deformation of breasts, deformation of breasts during activities and postural changes, and the effects of material properties and size of bra components on 3D breast deformation.

5. Conclusions

A non-linear FEM has been established to simulate human breast motion based on photogrammetric geometric models of 3D breast surfaces and biomechanical properties of different breasts tested by FEM and motion analysis. The model consisted of a thorax, two breasts and three skin layers with different sets of properties (density, Young's modulus, Poisson's ratio and shear modulus). The shear modulus of the different materials involved was estimated using FEM and validated with experimental data on breast motion. From the free breast vibration tests, the viscous damping ratio of breasts was ascertained to be 0.215 for the 80B breast size. The model was able to predict the breast displacement in which large deformation and non-linear material parameters were involved with good accuracy.

This study has demonstrated that potential benefits of FEM in the design/engineering of bras and for evaluating their performance in reducing breast motion from a theoretical perspective as opposed to the current empirical practices, thereby reducing the lead time for new product development. Furthermore, such an improved scientific understanding can provide guidelines for women to choose their bras with appropriate materials and pattern parameters for better fit, support and comfort for various situations such as breastfeeding, post-surgery or sports exercises such as yoga, jogging and dancing. This will bring long-term benefits to women's health and their families' well-being.

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